1S Whoop AIO FC - Datasheet reference circuits

Application / reference circuits pulled from each part's datasheet, with notes on what to copy.
Each part starts with a divider page; the pages after it are the original datasheet pages.

Flight controller

STM32G473CEU6 - Main MCU (U101) (pages 78, 116, 132)

Sensors

ICM-42688-P - Gyro / IMU (U201) (pages 20)

SPL06-001 - Barometer (U202) (pages 15)

W25Q128JV - Blackbox flash (U203) (pages 6)

Receiver

SX1281 - ELRS 2.4 GHz radio (U301) (pages 145, 147)

ESP8285 - ELRS RX MCU (U302) (pages 11, 12, 14, 15)

ESC

EFM8BB52 - Bluejay ESC MCU x4 (U401-U404) (pages 44)

Power

TPS63802 - 3.3 V buck-boost (U501) (pages 17)

TPS61023 - 5 V boost, FPV variant (U502) (pages 12)

Video

AT7456E - OSD (U603) (pages 5)

RTC6705 - 5.8 GHz VTX (U601) (no circuit in datasheet - notes only)

SE5004L - 5.8 GHz PA (U602) (no circuit in datasheet - notes only)

Connectors

USBLC6-2SC6 - USB ESD (U701) (pages 8)

STM32G473CEU6

Main MCU (U101) | Flight controller

Source: ST DS12712 STM32G473xB/xC/xE datasheet (via LCSC C1342773)

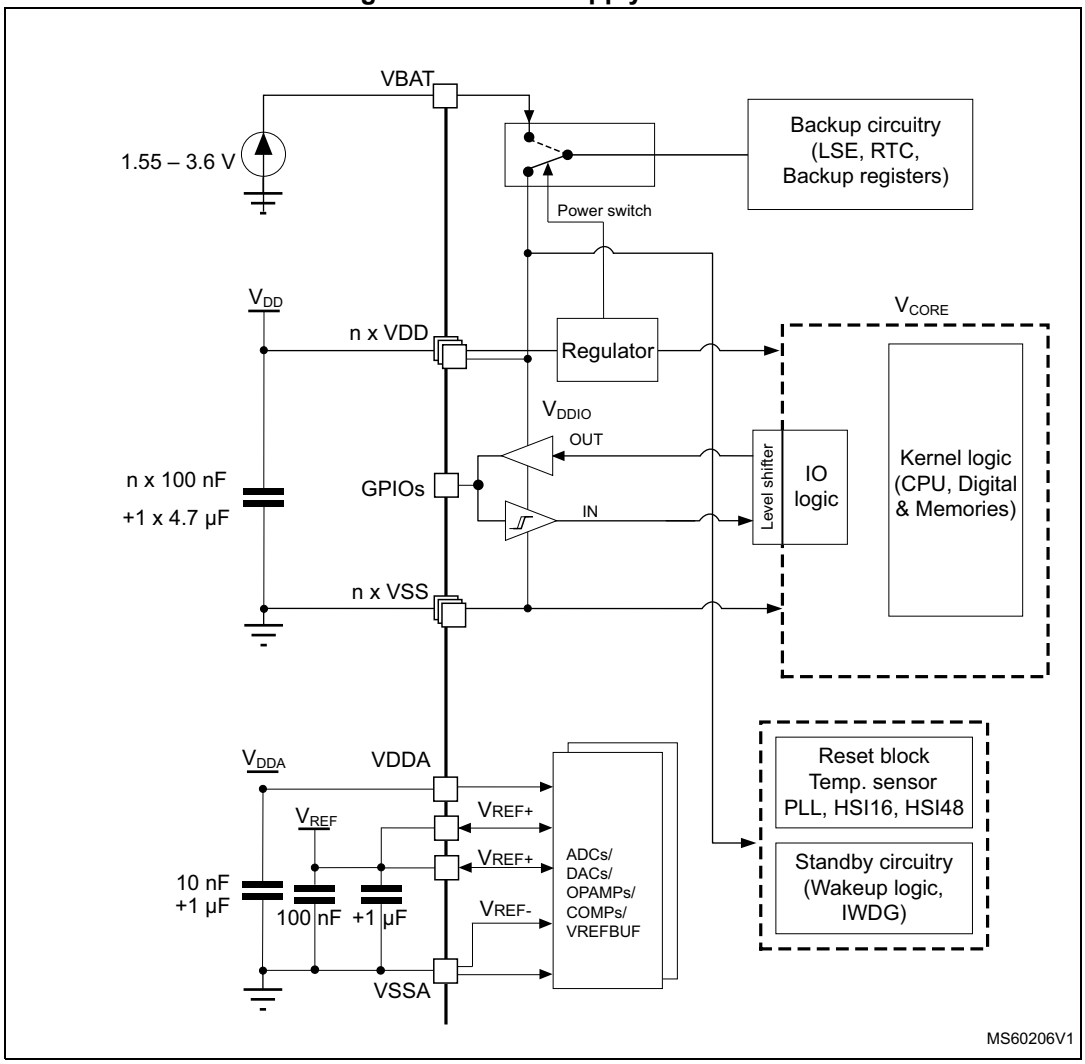
Datasheet pages included: 78, 116, 132

What to copy / watch for

- Fig 16 Power supply scheme: 100 nF on every VDD pin + one 4.7 uF bulk; VDDA gets 10 nF + 1 uF; VBAT to VDD with 100 nF if no backup battery.
- Fig 21 8 MHz HSE crystal: CL1/CL2 load caps sized from the crystal's CL (C_L = (CL1*CL2)/(CL1+CL2) + stray). REXT is optional.
- Fig 27 NRST: 100 nF to GND. Internal pull-up is already present.
- Not shown here: BOOT0 (PB8) 10k pull-down; USB D+/D- connect straight to PA12/PA11 (internal pull-up, no external resistors).
- Pin-map against an existing Betaflight STM32G47X target before wiring.

5.1.6 Power supply scheme

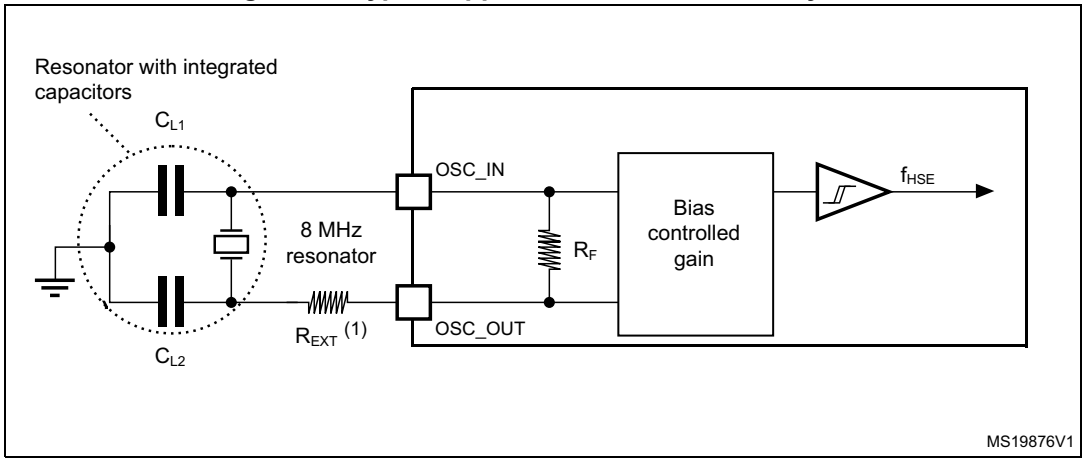
Figure 16. Power supply scheme



Caution: Each power supply pair (V_{DD}/V_{SS}, V_{DDA}/V_{SSA} etc.) must be decoupled with filtering ceramic capacitors as shown above. These capacitors must be placed as close as possible to, or below, the appropriate pins on the underside of the PCB to ensure the good functionality of the device.

Note: For information on selecting the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website www.st.com.

Figure 21. Typical application with an 8 MHz crystal

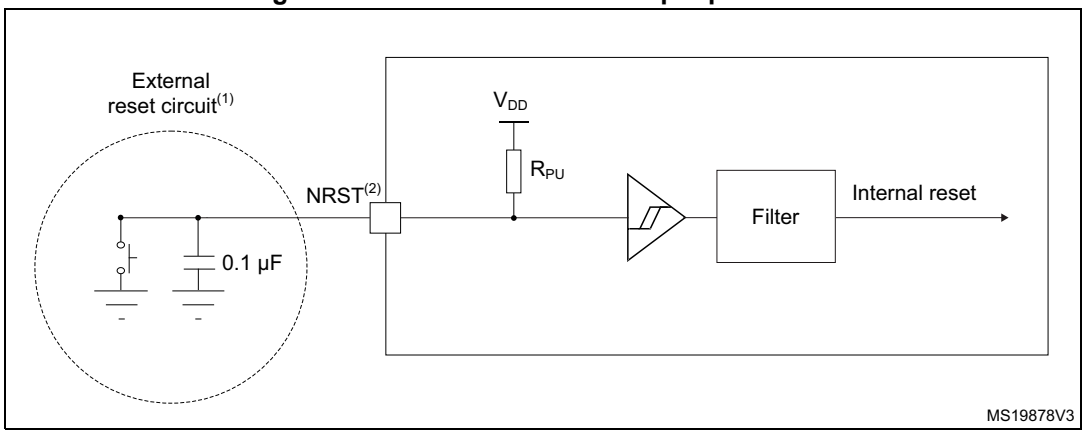


1. R_{EXT} value depends on the crystal characteristics.

Low-speed external clock generated from a crystal resonator

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal resonator oscillator. All the information given in this paragraph are based on design simulation results obtained with typical external components specified in *Table 42*. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Figure 27. Recommended NRST pin protection



1. The reset network protects the device against parasitic resets.
2. The user must ensure that the level on the NRST pin can go below the $V_{IL(NRST)}$ max level specified in *Table 58: NRST pin characteristics*. Otherwise the reset is not taken into account by the device.
3. The external capacitor on NRST must be placed as close as possible to the device.

5.3.16 Extended interrupt and event controller input (EXTI) characteristics

The pulse on the interrupt input must have a minimal length in order to guarantee that it is detected by the event controller.

Table 59. EXTI input characteristics(1)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
PLEC	Pulse length to event controller	-	20	-	-	ns

1. Guaranteed by design.

5.3.17 Analog switches booster

Table 60. Analog switches booster characteristics(1)

Symbol	Parameter	Min	Typ	Max	Unit
V_{DD}	Supply voltage	1.62	-	3.6	V
$t_{SU(BOOST)}$	Booster startup time	-	-	240	µs
$I_{DD(BOOST)}$	Booster consumption for 1.62 V ≤ V_{DD} ≤ 2.0 V	-	-	250	µA
	Booster consumption for 2.0 V ≤ V_{DD} ≤ 2.7 V	-	-	500	
	Booster consumption for 2.7 V ≤ V_{DD} ≤ 3.6 V	-	-	900	

1. Guaranteed by design.

ICM-42688-P

Gyro / IMU (U201) | Sensors

Source: TDK DS-000347 ICM-42688-P datasheet

Datasheet pages included: 20

What to copy / watch for

- Use Figure 6 (SPI interface). Figure 5 is I2C/I3C.
- VDD: C1 0.1 uF + C2 2.2 uF. VDDIO: 0.1 uF (C4 in figure).
- Pins 2, 3, 7, 10, 11 (RESV): GND or no-connect as shown. INT2/FSYNC (pin 9) to GND if unused.
- INT1 to an MCU EXTI pin for gyro sync. Give the gyro its own SPI bus.

4.2 TYPICAL OPERATING CIRCUIT

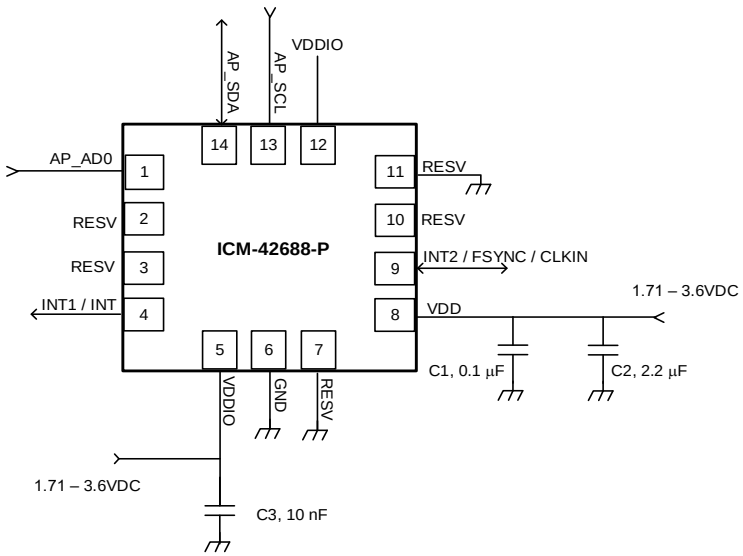


Figure 5. ICM-42688-P Application Schematic (I3C^{SM} / I^{2}C Interface to Host)

Note: I^{2}C lines are open drain and pull-up resistors (e.g. 10 kΩ) are required.

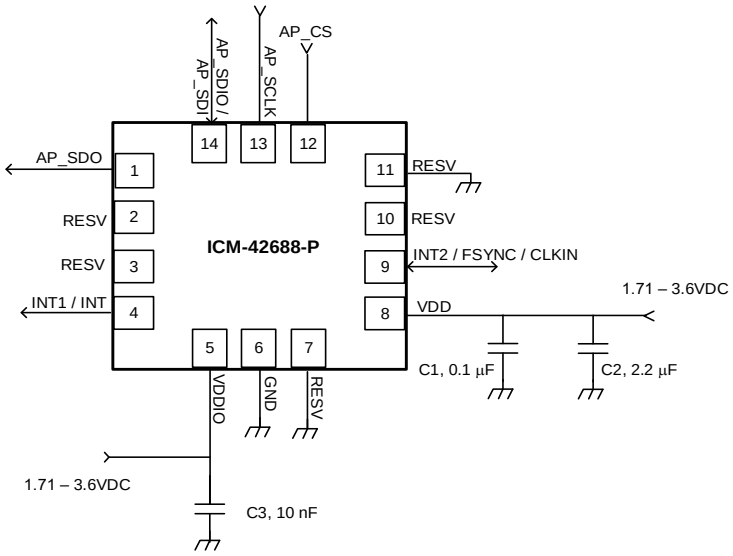


Figure 6. ICM-42688-P Application Schematic (SPI Interface to Host)

Document Number: DS-000347
Revision: 1.2

SPL06-001

Barometer (U202) | Sensors

Source: Goertek SPL06-001 datasheet v7.0 (via LCSC C2684428)

Datasheet pages included: 15

What to copy / watch for

- Figure 1 is I2C mode: CSB tied to VDDIO, SDO selects the address LSB, R1/R2 pull-ups on SDA/SCL.
- 100 nF on VDD and on VDDIO, close to the pins.
- DPS310 is pin-compatible if you swap parts.

6.2 Application Circuit Example

The example application circuit example uses the I2C serial interface. The SDO pin can be used for interrupt or to set least significant bit of the device address.

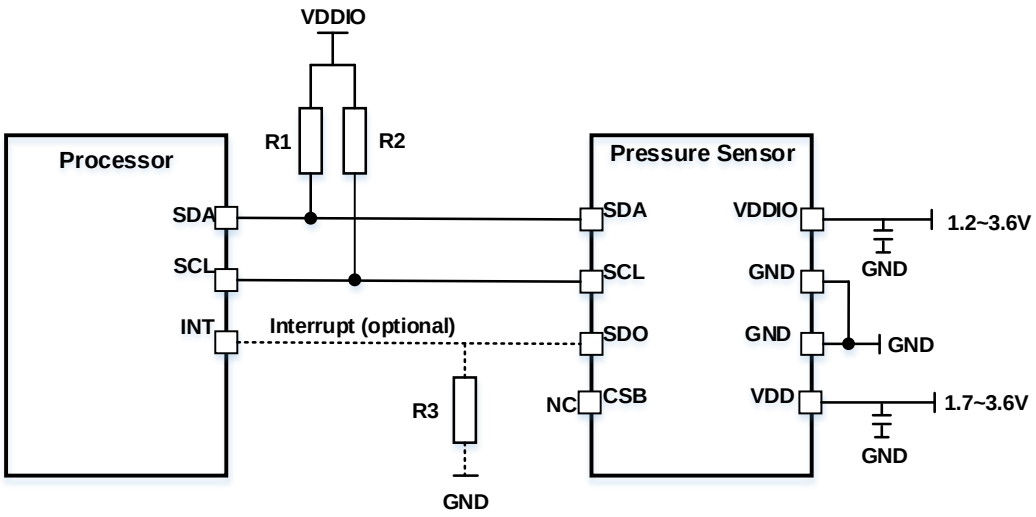


Figure 1: Typical application circuit

Table 6 Component Values

Component	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Pull-up/down Resistor	R_1, R_2			10	KΩ	
	R_3			100	KΩ	R_3 is optional and will set the address to 0x76 instead of 0x77.
Supply Blocking Capacitor	C_1, C_2	100	100		nF	The blocking capacitors should be placed as close to the package pins as possible.

W25Q128JV

Blackbox flash (U203) | Sensors

Source: Winbond W25Q128JV datasheet (via LCSC C97521)

Datasheet pages included: 6

What to copy / watch for

- The datasheet has no application circuit; this is the WSON-8 pinout / pin description.
- Single-SPI wiring: /CS, CLK, DI, DO to the shared SPI bus (with the OSD). /WP (IO2) and /HOLD (IO3) pull high to VCC.
- 100 nF on VCC; 10k pull-up on /CS so it stays deselected during MCU boot.

3. PACKAGE TYPES AND PIN CONFIGURATIONS

3.1 Pin Configuration SOIC 208-mil

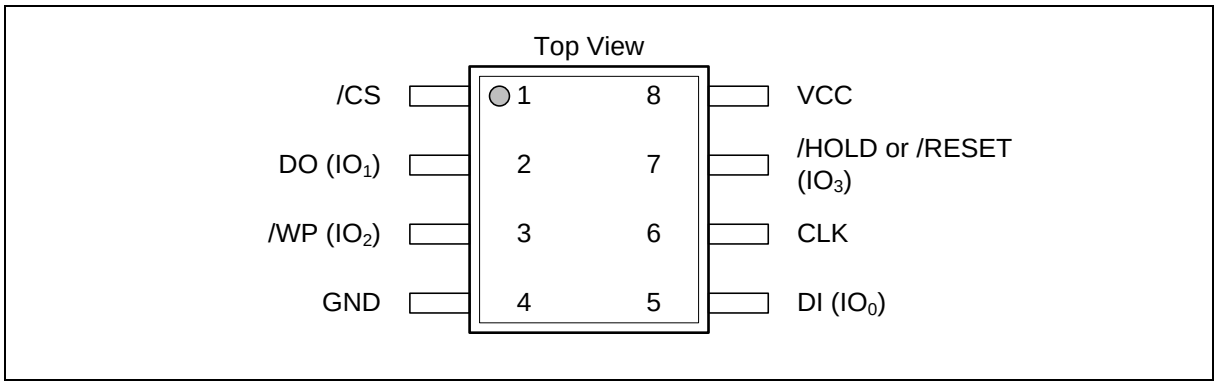


Figure 1a. W25Q128JV Pin Assignments, 8-pin SOIC 208-mil (Package Code S)

3.2 Pad Configuration WSON 6x5-mm/ 8x6-mm

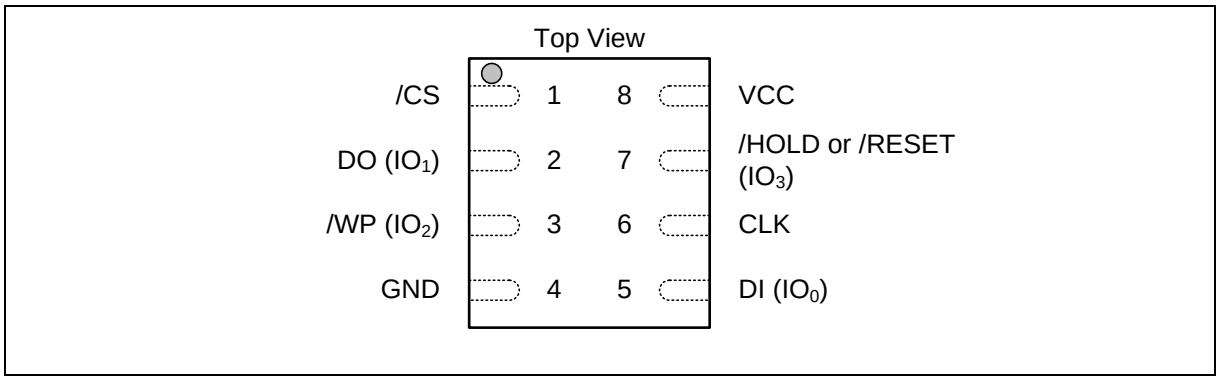


Figure 1b. W25Q128JV Pad Assignments, 8-pad WSON 6x5-mm/ 8x6-mm (Package Code P/E)

3.3 Pin Description SOIC 208-mil, WSON 6x5-mm / 8x6-mm

PAD NO.	PAD NAME	I/O	FUNCTION
1	/CS	I	Chip Select Input
2	DO (IO1)	I/O	Data Output (Data Input Output 1)(1)
3	/WP (IO2)	I/O	Write Protect Input (Data Input Output 2)(2)
4	GND		Ground
5	DI (IO0)	I/O	Data Input (Data Input Output 0)(1)
6	CLK	I	Serial Clock Input
7	/HOLD or /RESET (IO3)	I/O	Hold or Reset Input (Data Input Output 3)(2)
8	VCC		Power Supply

Notes:

1. IO0 and IO1 are used for Standard and Dual SPI instructions
2. IO0 – IO3 are used for Quad SPI instructions, /HOLD (or /RESET) function is only available for Standard/Dual SPI.

SX1281

ELRS 2.4 GHz radio (U301) | Receiver

Source: Semtech SX1280/SX1281 datasheet Rev 3.2, section 15 (via LCSC C2151551)

Datasheet pages included: 145, 147

What to copy / watch for

- Fig 15-1 is the reference design with the internal DC-DC (L between DCC_SW and DCC_FB).
- Fig 15-3 (TCXO variant) is what ELRS receivers use: 52 MHz TCXO AC-coupled into XTA, XTB open.
- Copy the RF match / harmonic filter values exactly; these are the 'TBD' L/C parts in the schematic.
- SPI (NSS, SCK, MOSI, MISO), BUSY, DIO1 and NRESET go to the ESP8285 (standalone RX) - see the ExpressLRS RX targets for GPIO mapping.

15. Reference Design and Application Schematics

15.1 Reference Design

15.1.1 Application Design Schematic

The long range 2.4 GHz application circuit is shown below:

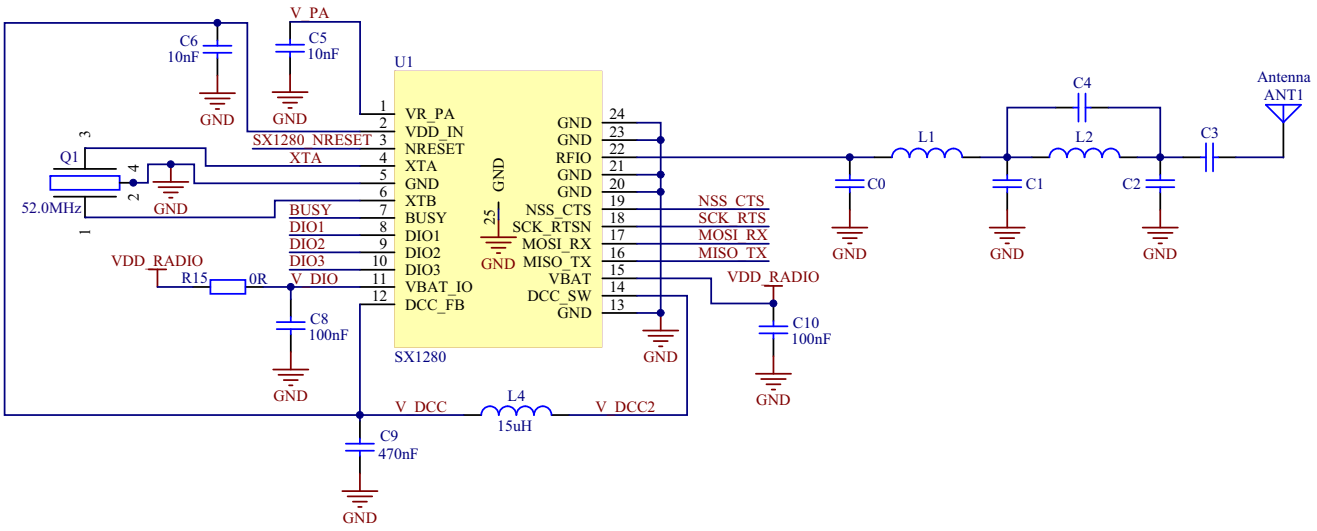


Figure 15-1: Transceiver Application Design Schematic

Figure 14-1 shows the SX1280 reference design. The design minimizes the number of external components needed to help reduce the overall cost and size of the design. The RF signal path comprises an impedance match (C0, L1) followed by a harmonic PI-section filter. This is then decoupled by C3 to allow the connection of grounded antennas to the antenna port without affecting the DC bias (applied by the circuit internally) on the RFIO pin.

C9 And L4 form the low pass filter for the integrated DC-DC regulation, the remaining capacitances are all related to decoupling of the battery and internally regulated supplies. Finally, the external crystal reference oscillator exploits internally integrated foot capacitances to further miniaturize the design.

15.2 Application Design with optional TCXO

Although the modulation parameters of the transceiver are designed to tolerate typical frequency drifts associated with the use of industry standard tolerance crystal reference oscillator components, an external Temperature Compensated Crystal Oscillator (TCXO) can be used. The figure below shows how the TCXO should be AC-coupled to the XTA input of transceiver. The TCXO should be of the 0.8 V clipped sine output type. Please consult the crystal manufacturer for full details of the components required specific to your TCXO.

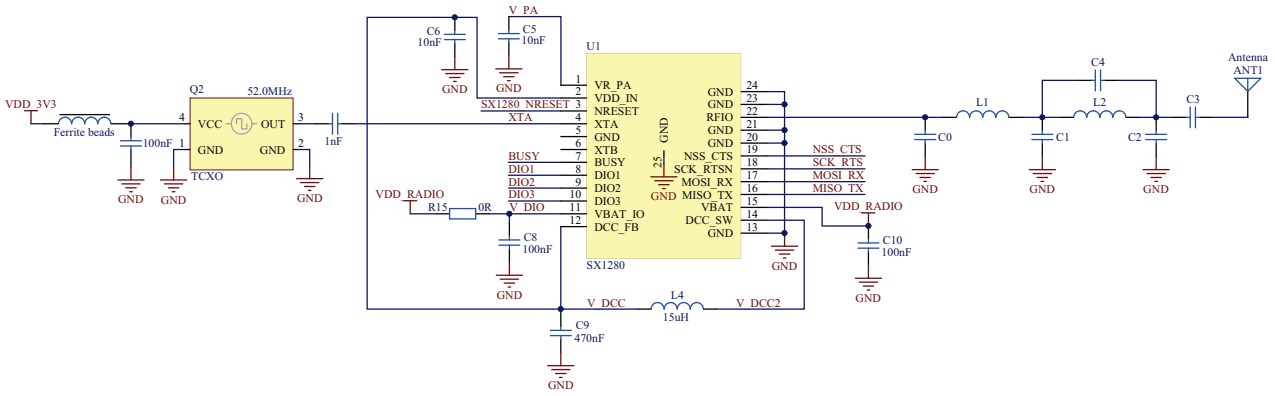


Figure 15-3: Application Schematic with Optional TCXO

15.3 Application Design with Low Drop Out Regulator

The following schematic shows the connection for use of the LDO instead of the DC-DC converter.

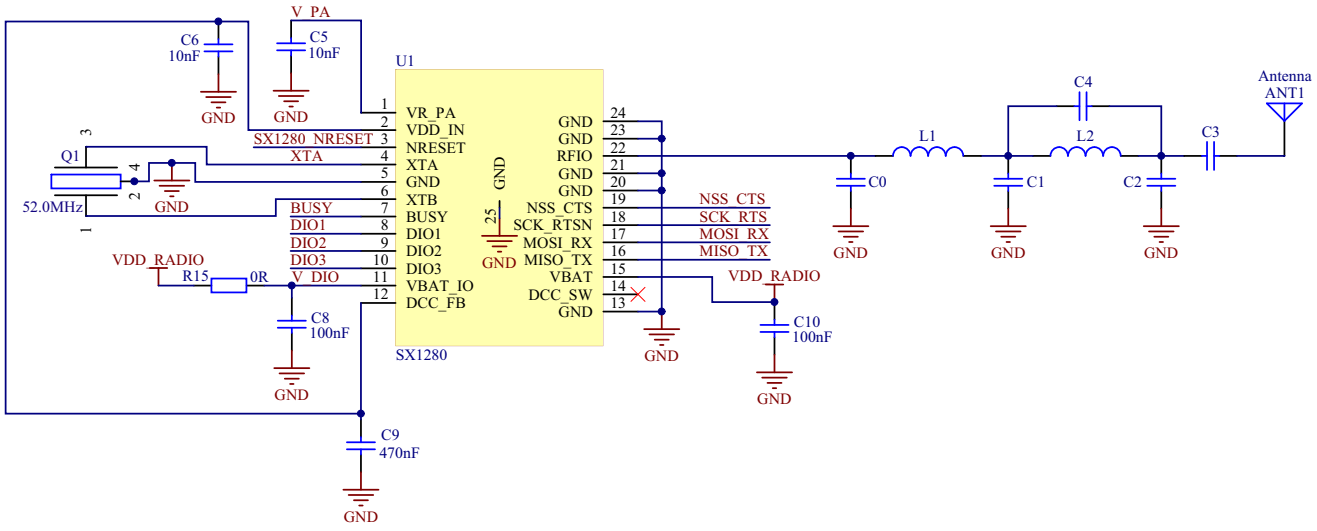


Figure 15-4: Application Schematic with Low Drop Out Regulator Schematic

ESP8285

ELRS RX MCU (U302) | Receiver

Source: Espressif ESP8266 Hardware Design Guidelines (ESP8285 is pin-compatible)

Datasheet pages included: 11, 12, 14, 15

What to copy / watch for

- The ESP8285 datasheet has no schematic; Espressif's ESP8266 guide applies pin-for-pin.
- IGNORE the external SPI flash in Fig 1-3/1-6: the ESP8285 has 1 MB flash inside, and pins 18-23 must stay unconnected to flash.
- Fig 1-4/1-5 power decoupling, Fig 1-7 26 MHz crystal, Fig 1-9 RES12K 12k 1% to GND.
- UART0 (U0TXD/U0RXD) carries CRSF to an STM32 UART. GPIO0 low at boot = bootloader (the BIND/BOOT pad).
- Fig 1-8 RF: the LNA pin is the Wi-Fi antenna port (used for ELRS Wi-Fi updates).

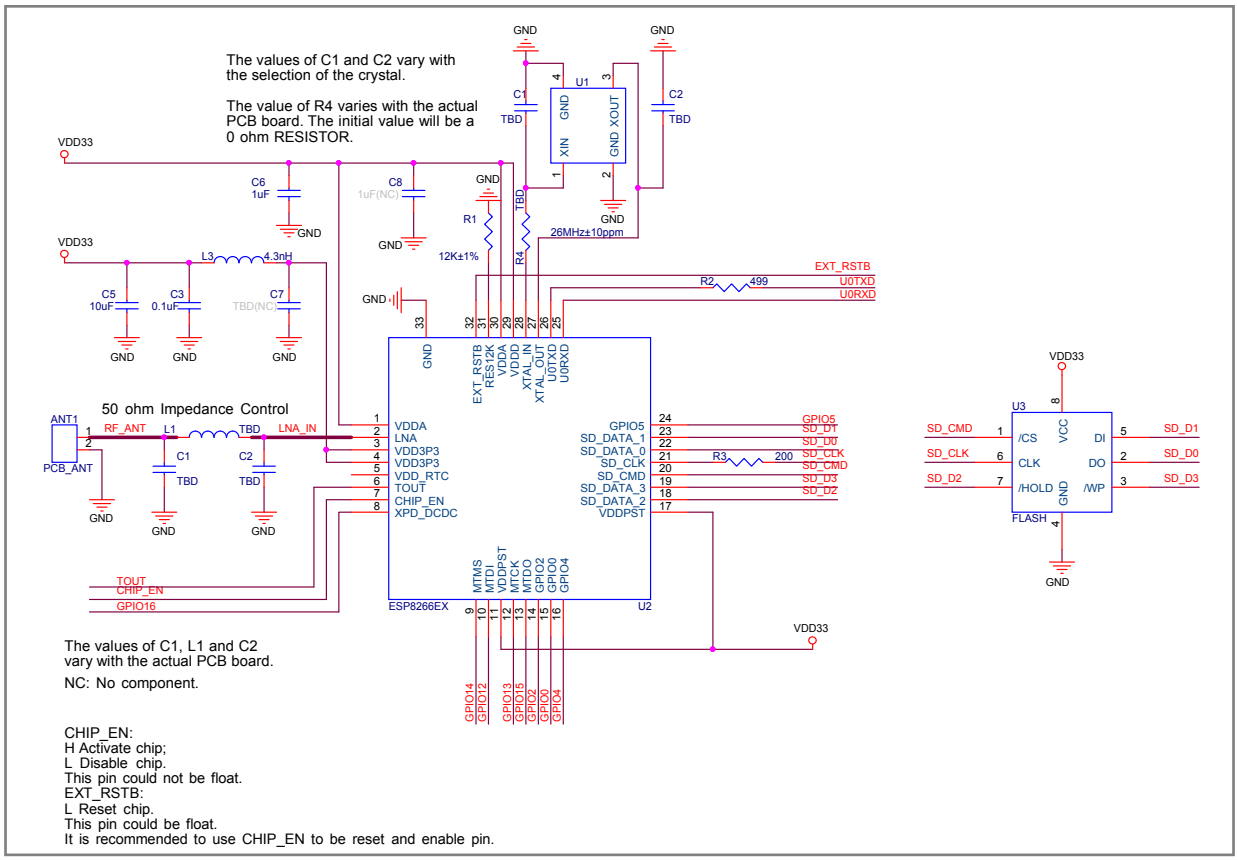


Figure 1-3. ESP8266EX Module Schematics

The ESP8266EX schematics include seven aspects:

- Power supply
- Power-on sequence and reset
- Flash
- Crystal oscillator
- RF
- External resistor
- UART

1.4.1. Power Supply

1.4.1.1. Digital Power Supply

ESP8266EX has two digital pins for power supply, Pin11 and Pin17. For digital power supply, there is no need to add additional filter capacitors. The operating voltage range of digital power supply pins is 1.8 V ~ 3.3 V.

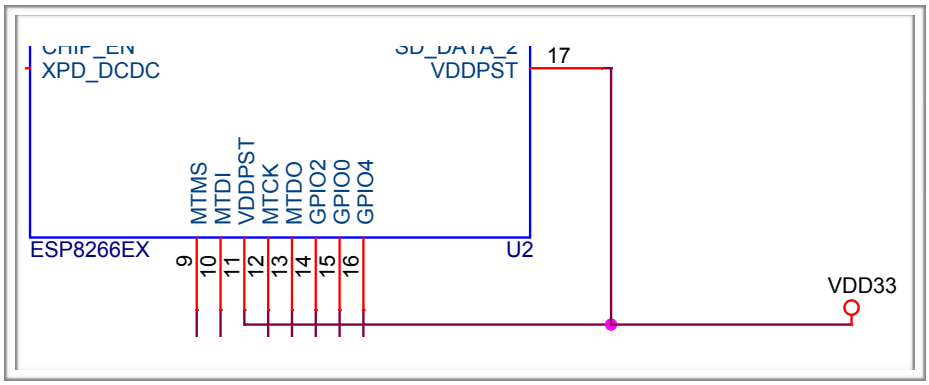


Figure 1-4. ESP8266EX Digital Power Supply Pins

1.4.1.2. Analog Power Supply

ESP8266EX has five analog pins for power supply, including Pin1, Pin3, Pin4 that are the power supply for internal PA and LNA; and Pin29, Pin30 for the internal PLL. The operating voltage for analog power supply pins is 2.5 V ~ 3.6 V.

Note that the power supply channel might be damaged due to the sudden increase of current when ESP8266EX is transmitting analog signals. Therefore, an additional 10 µF capacitor with a 0603 or 0805 package is needed to match the 0.1 µF capacitor.

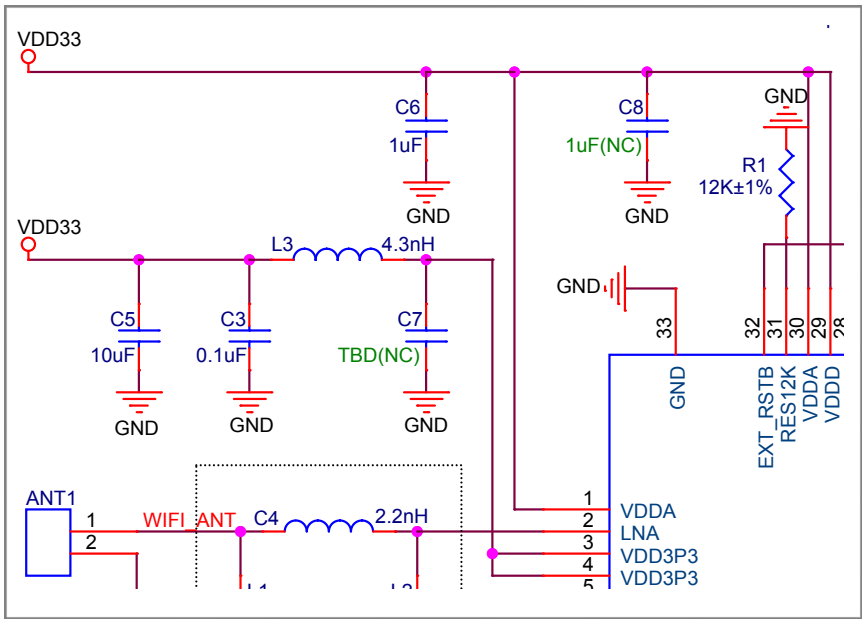


Figure 1-5. ESP8266EX AVDD

> 📖 **Note:**
>
> - *ESP8266EX's EMC is in conformity with FCC and CE requirements. There is no need to add ferrite beads in the analog power-supply circuit.*
> - *When using a single power supply, the recommended output current is 500 mA.*
> - *It is suggested that users add an ESD tube at the power entrance.*

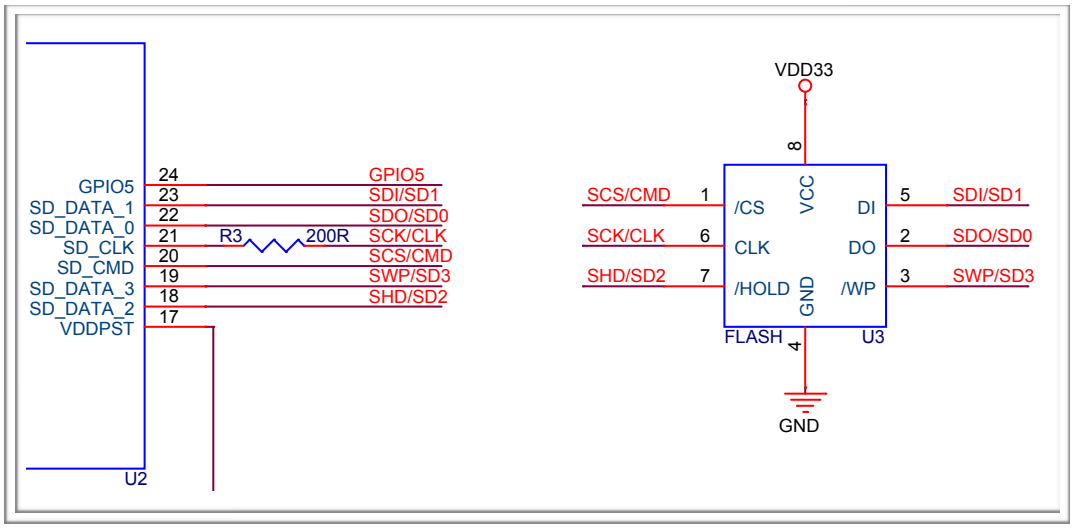


Figure 1-6. ESP8266EX Flash

1.4.4. Crystal Oscillator

ESP8266EX can support 40 MHz, 26 MHz and 24 MHz crystal oscillators.Please select the right type of crystal oscillator that is used in the ESP Flash Download Tool. In circuit design, capacitors C1 and C2, which are connected to the ground are added to the input and output terminals of the crystal oscillator respectively. The values of the two capacitors can be flexible, ranging from 6 pF to 22 pF. However, the specific capacitive values of C1 and C2 depend on further testing of and adjustment to the overall performance of the whole circuit. The crystal precision should be ±10 PPM.

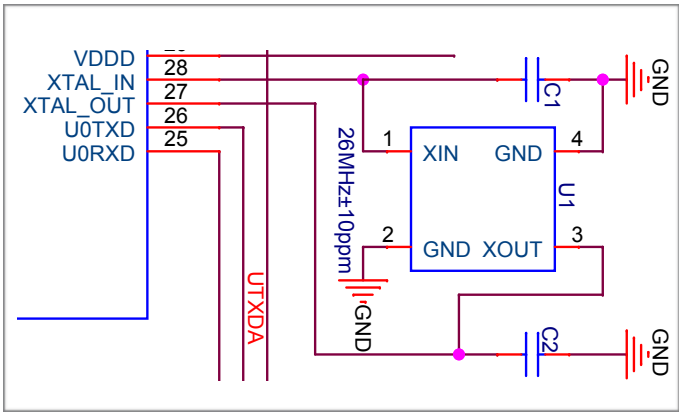


Figure 1-7. ESP8266EX Crystal Oscillator

> ⚠ **Notice:**
>
> *Defects in the craftsmanship of the crystal oscillators (for example, high frequency deviation and unstable working temperature) may lead to the malfunction of ESP8266EX, resulting in the decrease of overall performance.*

1.4.5. RF

The impedance of the ESP8266 PA output end is 39 + j6 Ω, so the matched impedance is 39-j6 Ω (from antenna to the chip).

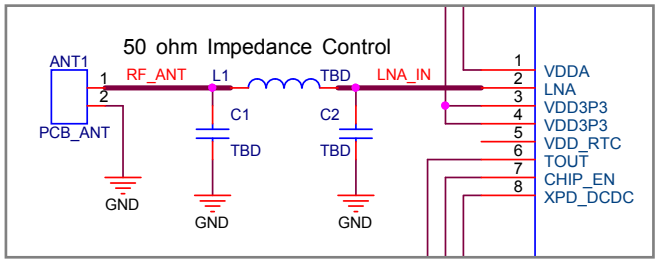


Figure 1-8. ESP8266EX RF

1.4.6. External Resistor 12K

An external ground resistor should be connected o the ERS12K pin (Pin31). The ground resistor requires high accuracy when controlling the bias current. An accuracy of 12K ± 1% is recommended.

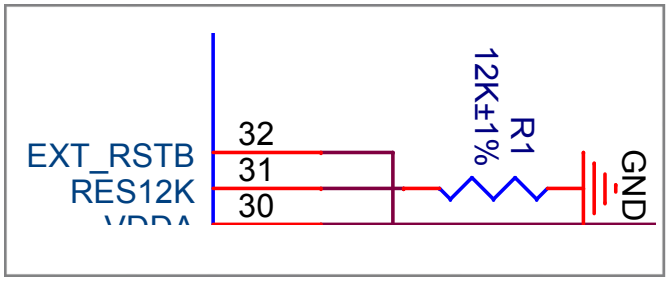


Figure 1-9. ESP8266EX External Resistor

1.4.7. UART

Users need to connect a 499 R resistor to the U0TXD line in order to suppress the 80 MHz harmonics. See Figure 1-10a.

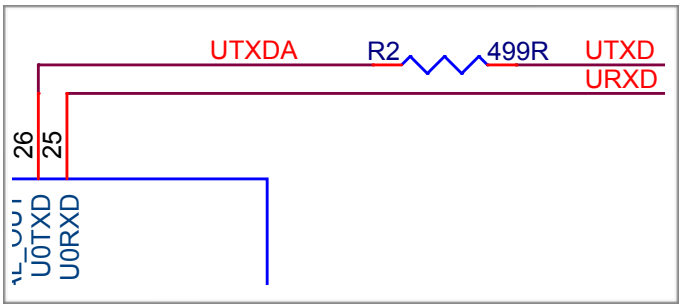


Figure 1-10a. ESP8266EX UART

By default, UART0 will output some printed information when ESP8266EX is powered on. For the applications that are sensitive to this feature, users can exchange the pins of UART (UART SWAP) during system initialization, that is, exchange U0TXD, U0RXD with U0RTS

EFM8BB52

Bluejay ESC MCU x4 (U401-U404) | ESC

Source: Silicon Labs EFM8BB52 datasheet Rev 1.0, section 5 (via LCSC C3225449)

Datasheet pages included: 44

What to copy / watch for

- Fig 5.1: 1 uF + 0.1 uF on VDD per MCU.
- Fig 5.2: C2 debug - 1k series resistors on C2CK (RSTb) and C2D, 1k pull-up on C2CK. Bring C2CK/C2D to the factory test pads.
- The FET gate drive, back-EMF dividers and neutral network are not in any datasheet. Copy them from a Bluejay-supported 1S ESC (e.g. the BB51/BB52 hardware targets).

5. Typical Connection Diagrams

5.1 Power

Figure 5.1 Power Connection Diagram on page 44 shows a typical connection diagram for the power pins of the EFM8BB52 devices.

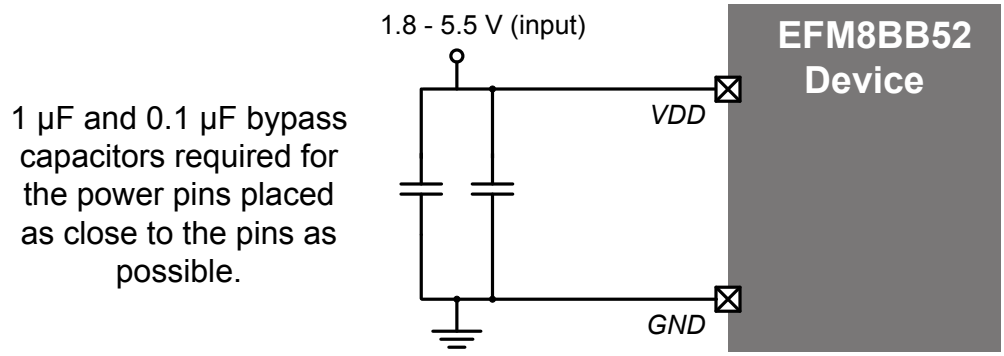


Figure 5.1. Power Connection Diagram

5.2 Debug

The diagram below shows a typical connection diagram for the debug connections pins. The pin sharing resistors are only required if the functionality on the C2D (a GPIO pin) and the C2CK (RSTb) is routed to external circuitry. For example, if the RSTb pin is connected to an external switch with debouncing filter or if the GPIO sharing with the C2D pin is connected to an external circuit, the pin sharing resistors and connections to the debug adapter must be placed on the hardware. Otherwise, these components and connections can be omitted.

For more information on debug connections, see the example schematics and information available in application note, *AN124: Pin Sharing Techniques for the C2 Interface*. Application notes can be found on the Silicon Labs website (http://www.silabs.com/8bit-appnotes) or in Simplicity Studio.

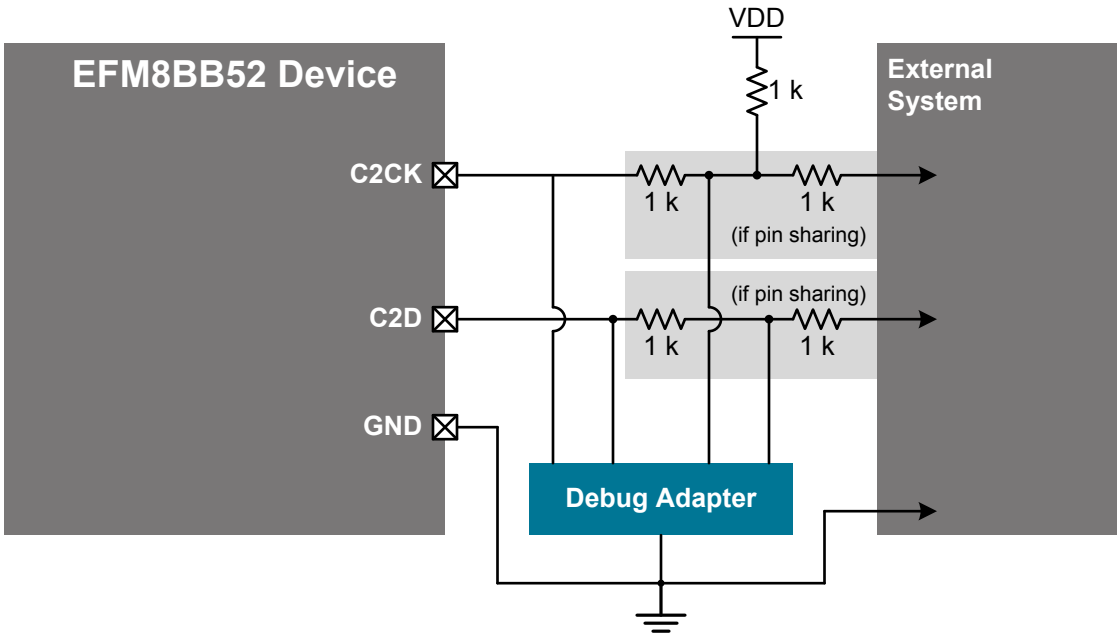


Figure 5.2. Debug Connection Diagram

TPS63802

3.3 V buck-boost (U501) | Power

Source: TI SLVSEU9D TPS63802 datasheet

Datasheet pages included: 17

What to copy / watch for

- Fig 10-1: L1 0.47 uH, C1 10 uF in, C2 22 uF out (the schematic has 2 x 22 uF for extra margin).
- EN tied to VIN, MODE to GND (power-save) or VIN (forced PWM, lower ripple for the gyro). PG pull-up R3 100k is optional.
- FB divider R1/R2: Vout = 0.5 V x (1 + R1/R2); 560k/100k gives 3.3 V.

10 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

10.1 Application Information

The TPS63802 is a high efficiency, low quiescent current, non-inverting buck-boost converter, suitable for applications that need a regulated output voltage from an input supply that can be higher or lower than the output voltage.

10.2 Typical Application

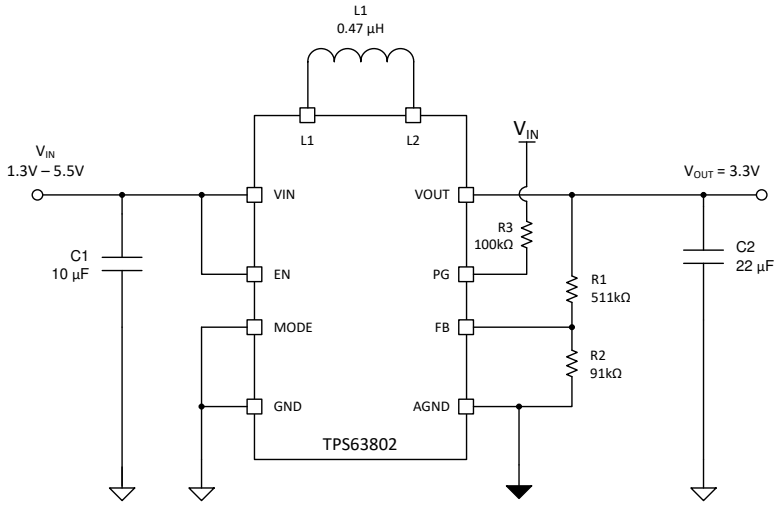


Figure 10-1. 3.3 V_{OUT} Typical Application

10.2.1 Design Requirements

The design guideline provides a component selection to operate the device within *Table 10-1*.

Table 10-1 shows the list of components for the application characteristic curves.

Table 10-1. Matrix of Output Capacitor and Inductor Combinations

NOMINAL INDUCTOR VALUE [µH](1)	NOMINAL OUTPUT CAPACITOR VALUE [µF](2)				
	10	22	47	66	100
0.47	-	+ (3)	+	+	+

(1) Inductor tolerance and current derating is anticipated. The effective inductance can vary by 20% and –30%.
(2) Capacitance tolerance and DC bias voltage derating is anticipated. The effective capacitance can vary by 20% and –50%.
(3) TPS63802 typical application. Other check marks indicate possible filter combinations.

10.2.2 Detailed Design Procedure

The first step is the selection of the output filter components. To simplify this process, *Section 8.1* outlines minimum and maximum values for inductance and capacitance. Take tolerance and derating into account when selecting nominal inductance and capacitance.

10.2.2.1 Custom Design With WEBENCH® Tools

Click here to create a custom design using the TPS63802 device with the WEBENCH® Power Designer.

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TPS61023

5 V boost, FPV variant (U502) | Power

Source: TI SLVSF14B TPS61023 datasheet

Datasheet pages included: 12

What to copy / watch for

- Fig 8-1: L1 1 uH, C1 10 uF, C2 2 x 22 uF, R1 732k / R2 100k for 5 V.
- The schematic now matches (2 x 22 uF output, 732k 1% / 100k).
- EN high to enable; drive it from an MCU pin if you want the 5 V rail switchable.

8 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

8.1 Application Information

The TPS61023 is a synchronous boost converter designed to operate from an input voltage supply range between 0.5 V and 5.5 V with a typically 3.7-A valley switch current limit. The TPS61023 typically operates at a quasi-constant 1-MHz frequency PWM at moderate-to-heavy load currents when the input voltage is above 1.5 V. The switching frequency changes to 0.5 MHz gradually with the input voltage changing from 1.5 V to 1 V for better efficiency and high step-up ratio. When the input voltage is below 1 V, the switching frequency is fixed at a quasi-constant 0.5 MHz. At light load currents, the TPS61023 converter operates in power-save mode with PFM to achieve high efficiency over the entire load current range.

8.2 Typical Application

The TPS61023 provides a power supply solution for portable devices powered by batteries or backup applications powered by super-capacitors. With typical 3.7-A switch current capability, the TPS61023 can output 5 V and 1.5 A from a single-cell Li-ion battery.

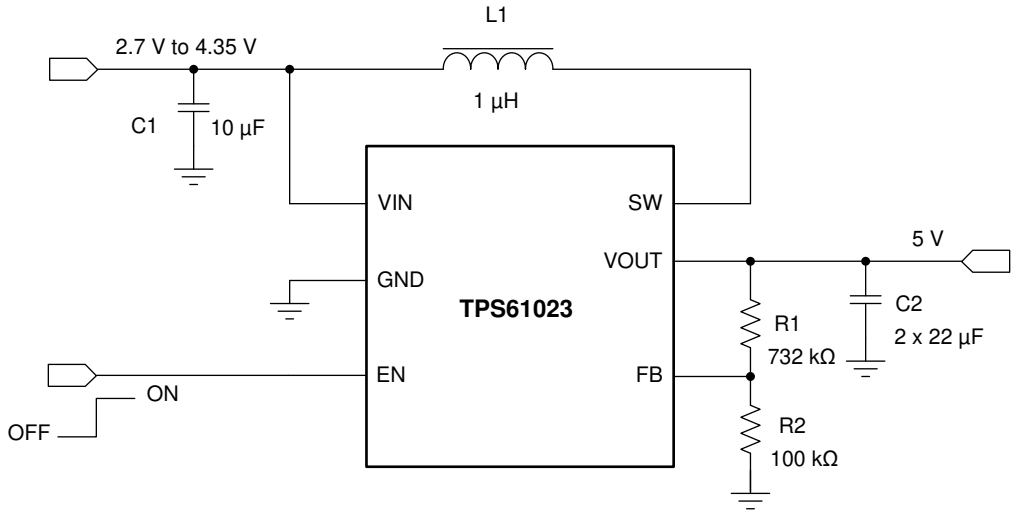


Figure 8-1. Li-ion Battery to 5-V Boost Converter

8.2.1 Design Requirements

The design parameters are listed in Table 8-1.

Table 8-1. Design Parameters

PARAMETERS	VALUES
Input voltage	2.7 V to 4.35 V
Output voltage	5 V
Output current	1.5 A
Output voltage ripple	±50 mV

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AT7456E

OSD (U603) | Video

Source: AT7456E-S402 datasheet V1.1 (via LCSC C82351)

Datasheet pages included: 5

What to copy / watch for

- Fig 2 (bottom of page): typical operating circuit. The upper drawing is the TSSOP-28 part used in the schematic.
- 27 MHz crystal between CLKIN and XFB with load caps; 0.1 uF on DVDD, AVDD and PVDD.
- Camera video into VIN through a 0.1 uF coupling cap with 75 ohm termination; VOUT goes to the VTX video input. SAG ties to VOUT if unused.
- Supply range is 3.15-5.25 V (electrical characteristics table), so running AVDD/DVDD/PVDD from the 3.3 V buck-boost rail is within spec.

Note 1: See the standard test circuits of Figure 1. RL = 75 , unless otherwise specified. All digital input signals are timed from a voltage level of (VIH + VIL) / 2. All parameters are tested at TA = +85°C and values through temperature range are guaranteed by design.

Note 2: The input-voltage operating range is the input range over which the output signal parameters are guaranteed (Figure 3).

Note 3: The input-voltage sync detection range is the input composite video range over which an input sync signal is properly detected and the OSD signal appears at VOUT. However, the output voltage specifications are not guaranteed for input signals exceeding the maximum specified in the input operating voltage range (Figure 3).

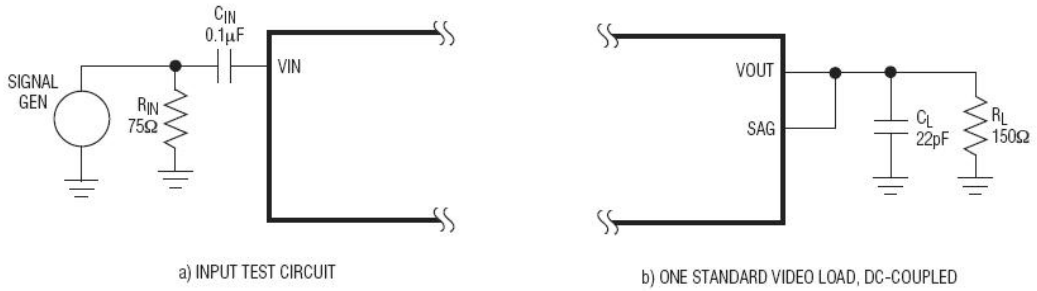
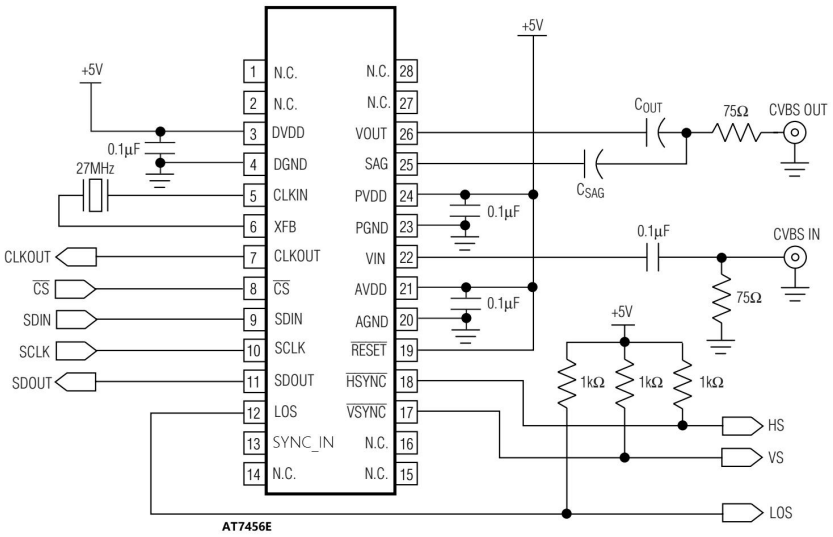


图 1. 标准测试电路



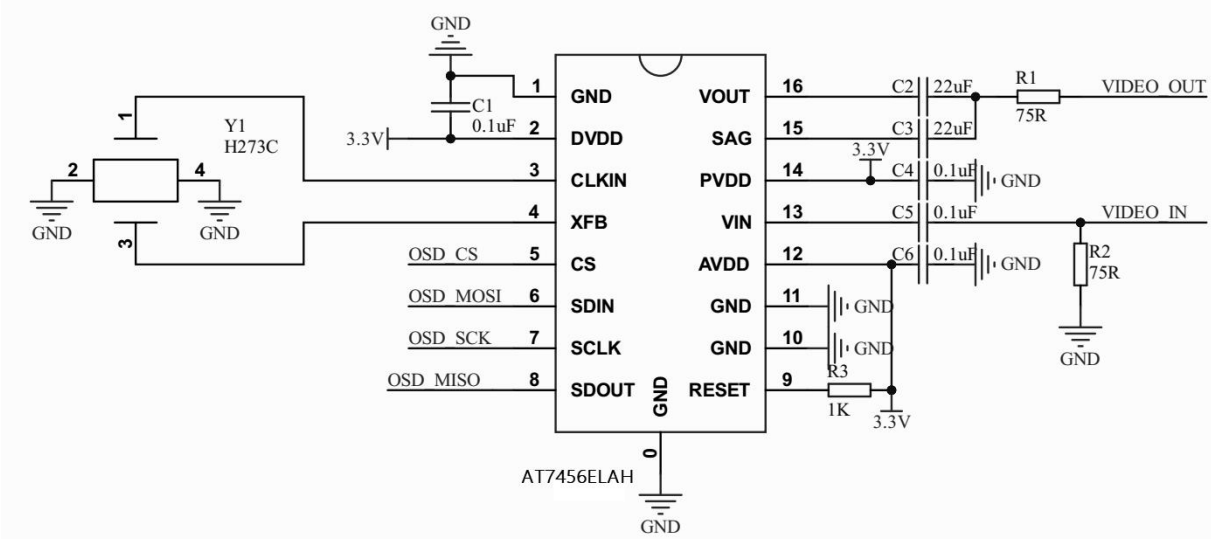


图 2. 典型工作电路

RTC6705

5.8 GHz VTX (U601) | Video

Source: RichWave RTC6705-DST-001 (via LCSC C913074)

Datasheet pages included: none

What to copy / watch for

- The RTC6705 datasheet contains NO application circuit - only pinout, registers and package.
- Copy the RTC6705 + PA section from a known-good whoop AIO or an open-source VTX design.
- From the datasheet: 8 MHz crystal on XTAL1/XTAL2; SPI_SE (pin 1) high selects SPI mode; PAOUT1 (pin 35) is the +13 dBm output.

SE5004L

5.8 GHz PA (U602) | Video

Source: Skyworks SE5004L datasheet 202393B

Datasheet pages included: none

What to copy / watch for

- The SE5004L datasheet has NO application schematic; the circuit is on the SE5004L-EK1 evaluation board.
- From the datasheet: VCC1/VCC2/VCC3 3.0-5.5 V, VREF 2.8-2.9 V to enable (0 V = off), about 6 decoupling caps, DC-blocking caps on RFIN and RFOUT.
- The schematic uses a WhoopFC symbol with the exposed pad as pin 21 (GND), so it gets wired to ground.

USBLC6-2SC6

USB ESD (U701) | Connectors

Source: ST USBLC6-2 datasheet (via LCSC C7519)

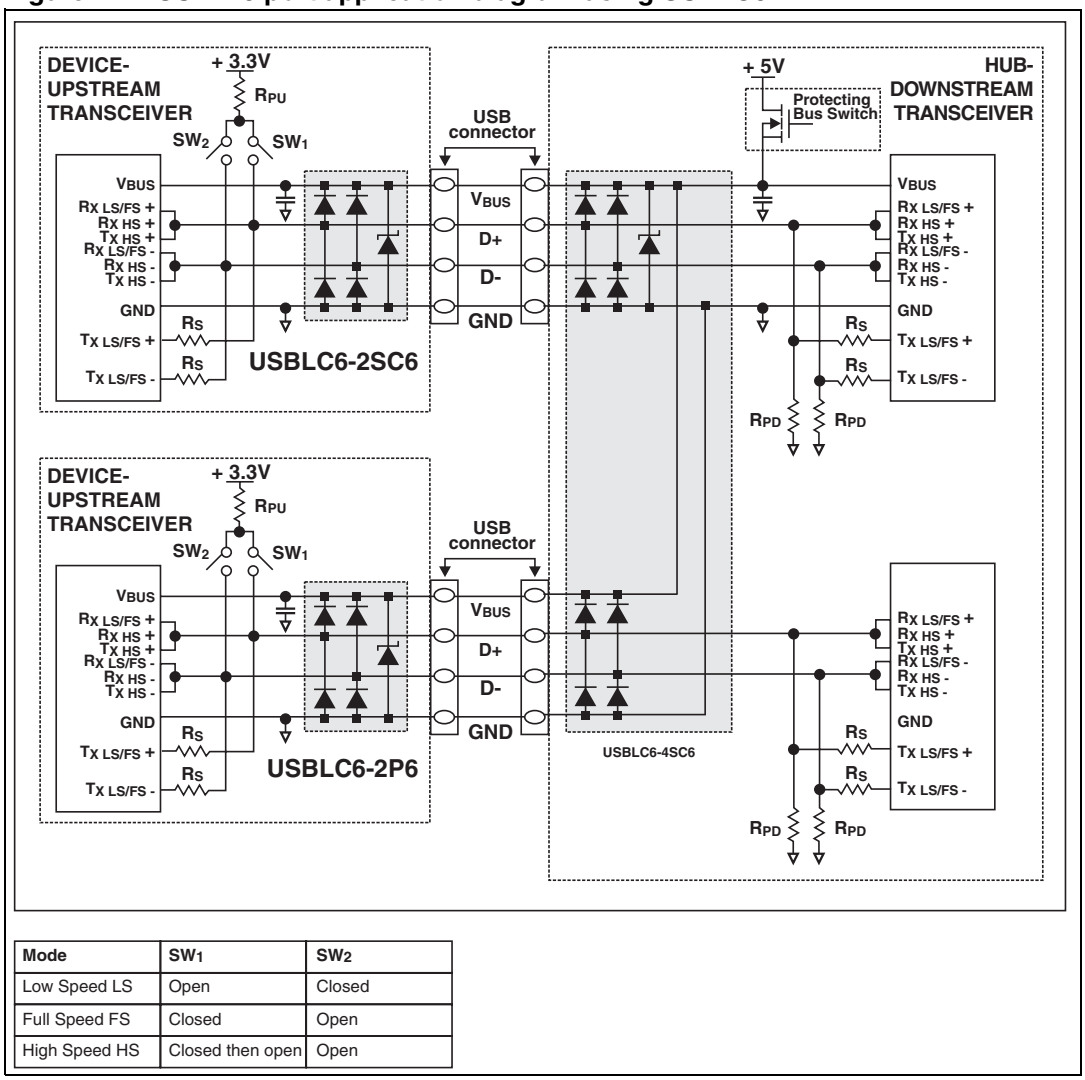
Datasheet pages included: 8

What to copy / watch for

- Fig 14 (top): USB 2.0 port. D+/D- pass through I/O1/I/O2 pairs; VBUS on pin 5, GND on pin 2.
- Place it right at the JST-SH config connector.

2.5 Application examples

Figure 14. USB 2.0 port application diagram using USBLC6-2



Mode	SW_1	SW_2
Low Speed LS	Open	Closed
Full Speed FS	Closed	Open
High Speed HS	Closed then open	Open

Figure 15. T1/E1/Ethernet protection

